

PROG2053Autumn 2026- Assignment 1

Assignment Overview

This assignment contains basic static web development concepts. Submissions are due on September **06th**, **2026**.

AI tools may be used for learning support only; please mention whether you have used them and for what purpose. All submitted work must be completed independently. This assignment carries specific weight toward your final overall grade.

Instructions

If any part of this assignment is unclear, students are expected to make reasonable assumptions and clearly document them within their submission (e.g., as comments in the code or in an accompanying note). All stated assumptions will be considered during the marking process.

Please upload the assignment as **' .txt' file**.

1: HTML Fundamentals & Form Implementation

Write an HTML code for the following web page as described below.
Use proper HTML structure and formatting.

Page Description:

1. The web page should have the title **“My Hobby Page”**.
2. Display a main heading **“My Favorite Hobbies”** at the top.
3. Below the heading, include a short paragraph:
“I enjoy spending my free time doing different activities that make me happy.”
4. Add an image named `hobby.jpg` with alternate text “Hobby Image”.
5. Create an unordered list of three hobbies: **Reading**, **Painting**, and **Traveling**.
6. Insert a horizontal line after the list.
7. Create a table with two columns — **Hobby** and **Time Spent per Week (hours)** — and fill it with three rows of your hobbies.
8. Include a link that says **“Learn more about hobbies”**, linking to <https://example.com/hobbies>.
9. Create another page named `contact.html` give its link at the bottom.
10. In `contact.html`, build a user inquiry form detailing how forms transmit data (using action and method attributes). Include the following form controls:

- **Text input** `<input type="text">` for user's name
- **Dropdown selector** `<select>` and `<option>` for inquiry types – Hobby Suggestions and Website Feedback
- **Textarea** `<textarea>` for detailed messages
- **Submit button** `<button type="submit">`
- **A link to go back** to 'My Hobby Page'

2: CSS Styling and Box Model

Part 1:

Suppose you are designing a promotional highlight card for a local community tech event. You are handed a structural HTML page containing a raw text framework inside a generic div layout box.

Your job is to transform this unstyled markup step-by-step into a premium, eye-catching event block entirely within the internal `<style>` block using separate CSS design properties.

```
<body>

  <div id="event-card" class="align-wrapper">

    <h2 class="event-title">AI & Design Meetup</h2>
    <p class="status-alert">Status: Registration Open</p>

  </div>

</body>
```

You are forbidden from modifying any of the HTML code down in the body.

Complete the following **6 diagnostic design tasks** inside your internal stylesheet:

- **Task 1: Universal Selector**
Target the universal selector `*`. Remove all browser-default margins and padding by setting `margin: 0; padding: 0;`.
- **Task 2: Element Selector & CSS Fonts**
Target the structural body tag directly. Use **CSS fonts** to enforce a clean, modern styling using `font-family: 'Helvetica Neue', Helvetica, sans-serif;`.
- **Task 3: ID Selector & CSS Background**
Target the unique ID (`#event-card`). Give it a soft light-green background color using `background-color: #f0fdf4;`. Set its width to 450px, add 25px padding inside and center the entire card on the screen using `margin: 60px auto;`.
- **Task 4: Class Selector & CSS Text Align**
Target the class identifier `.align-wrapper`. Use CSS text alignment to center all the text lines neatly inside the card layout using `text-align: center;`.

- **Task 5: Class Selector, CSS Color & CSS Fonts**
Target the class selector .event-title. Change its color to a dark charcoal tone using color: #1e293b; and and make the font size exactly 28px.
- **Task 6: Class Selector, CSS Color & CSS Text Decoration**
Target the sub-text class identifier .status-alert. Change its text color to an emerald green color using color: #059669;. Finally, add a wavy underline effect beneath the words using text-decoration: underline double #059669;.

Part 2:

1. What is the relative unit in CSS?
 - a) Explain what each CSS unit refers to (i.e., what it is relative to):
 - em
 - rem
 - %
 - vh
 - vw
 - vmin
 - vmax
 - b) Calculate the pixel values given:
 - Root (<html>) font-size = 18px
 - Parent element font-size = 22px
 - Parent container width = 500px
 - Viewport size = 1440px × 900px

○ Unit	○ CSS Rule
○ em	○ font-size: 1.8em;
○ rem	○ font-size: 2.5rem;
○ %	○ width: 60%;
○ vh	○ height: 7vh;
○ vw	○ width: 8vw;
○ vmin	○ width: 6vmin;
○ vmax	○ width: 6vmax;

2. **Explain the difference** between these two boxes and determine what happens to **total box size of box1 and box2** when you increase padding from 20px to 30px.

```
CSS

.box1 {
  width: 200px;
  padding: 20px;
  border: 5px solid;
  box-sizing: content-box;
}

.box2 {
  width: 200px;
  padding: 20px;
  border: 5px solid;
  box-sizing: border-box;
}
```

3. HTML-CSS Styling List, Table and Forms

1. Read each story. Decide which CSS property should be applied and write the code.

- **Scenario 1** : A cafe website needs lists. They want the bullets look like squares for the menu list.

```
<ul class="cafe-menu">
  <li>Hot Coffee</li>
  <li>Iced Tea</li>
  <li>Fresh Juice</li>
</ul>
```
- **Scenario 2** : A teacher makes a quiz. It needs numbers. Use small Roman numerals like i, ii, and iii.

```
<ol class="history-quiz">
  <li>Who discovered America?</li>
  <li>In what year did World War II end?</li>
  <li>What was the ancient capital of Egypt?</li>
</ol>
```
- **Scenario 3** : A store wants custom lists. They have an image named heart.png. Use this image as the bullet.

```
<ul class="game-features">
  <li>Retro Pixel Art</li>
  <li>Multiplayer Mode</li>
  <li>Custom Avatars</li>
</ul>
```
- **Scenario 4** : Text looks messy on phones. Move the bullets inside the text box block.

```
<ul class="mobile-list">
  <li>This is a very long sentence that wraps down to a second line on small phone
  screens.</li>
  <li>Another long sentence to test how the text behaves when it runs out of space.</li>
</ul>
```

- **Scenario 5:** The text looks too crowded. Add space between list items. Make the lines taller.

```
<ul class="readable-list">
  <li>First important news update of the morning.</li>
  <li>Second major announcement from the team.</li>
  <li>Third breaking update regarding the event.</li>
</ul>
```

2. Consider the following scenarios:

- **Scenario 1:** A sports team stats sheet has 100 rows, making it hard to track individual player metrics. To fix this, they want every single **alternate even row** to automatically change its background to a **light gray color (#eee)**.

```
<table class="stats-table">
  <tr>
    <th>Player</th>
    <th>Score</th>
  </tr>
  <tr>
    <td>Alex</td>
    <td>14</td>
  </tr>
  <tr class="even">
    <td>Blake</td>
    <td>22</td>
  </tr>
  <tr>
    <td>Charlie</td>
    <td>9</td>
  </tr>
  <tr class="even">
    <td>Dana</td>
    <td>31</td>
  </tr>
</table>
```

- **Scenario 2:** A user is looking at a huge data chart. To help them see which row they are looking at, make the background color of a row change **when the computer mouse hovers over it**. Also, if a box inside the chart is completely **empty**, **hide its borders**.

```
<table class="data-grid">
```

```
  <tr>
```

```
    <th>ID</th>
```

```
    <th>Status</th>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>001</td>
```

```
    <td>Active</td>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>002</td>
```

```
    <td></td>
```

```
  </tr>
```

```
</table>
```

- **Scenario 3:** A user clicks inside a text box to start typing an email address. To show the box is active and currently **receives focus (typing mode)**, change its background color to **white**.

```
<input type="email" class="email-box">
```

- **Scenario 4:** An app built for elderly users needs text fields that are very easy to see. Make the text inside the input boxes **large (120% size)** and change the color of the text to a distinct gray color (#555).

```
<input type="text" class="large-box">
```